



DAY 4 – PYTHON BOOTCAMP - WINTER EDITION 2026

Core Topic: Loops in Python

Resource: <https://www.youtube.com/watch?v=UrsmFxEIp5k> from 3hr 58 minutes till 4hr 50 minutes

[Expected Time to Finish - 2-3 hours]

Core Problem: Daily Expense Tracker

Everyone works on the same problem. Difficulty depends on how deep you take it. Choose your level and complete it properly.

BEGINNER LEVEL (Starting from Scratch)

Write a Python program that:

1. Asks the user how many days they want to track expenses for.
2. Uses a **loop** to take daily expense amounts from the user.
3. Stores all expenses in a **list**.
4. After the loop:
 - Print all expenses
 - Print the **total expense**
 - Print the **average expense**

Sample Input:

```
Enter number of days: 3
Enter expense for day 1: 200
Enter expense for day 2: 150
Enter expense for day 3: 250
```

Sample Output:

```
Expenses: [200, 150, 250]
Total Expense: 600
Average Expense: 200.0
```

INTERMEDIATE LEVEL

Smart Expense Analyzer with Alerts

● Intermediate Task (Upgraded)

Problem Statement:

Write a Python program that:

1. Asks the user how many expenses they want to enter.
2. Uses a loop to take:
 - Expense amount
 - Expense category (Food, Travel, Shopping, Other)
3. Stores:
 - All expense amounts in a list
 - Category-wise totals in a dictionary
4. During input, check:
 - If any single expense is greater than ₹500, print

“High expense detected”

5. After all inputs:
 - Calculate and print total expense
 - Calculate and print average expense
 - Print category-wise totals
 - Find and print the highest spending category
6. Use if / elif / else to print a spending status:
 - $\text{Average} \leq ₹200 \rightarrow \text{Controlled Spending}$
 - $\text{Average} \leq ₹400 \rightarrow \text{Moderate Spending}$
 - $\text{Average} > ₹400 \rightarrow \text{High Spending}$

Sample Input:

Enter number of expenses: 5

Expense 1 amount: 150

Expense 1 category: Food

Expense 2 amount: 600

Expense 2 category: Shopping

Expense 3 amount: 200

Expense 3 category: Travel

Expense 4 amount: 450

Expense 4 category: Food

Expense 5 amount: 300

Expense 5 category: Other

Sample Output:

High expense detected

Total Expense: 1700

Average Expense: 340.0

Category-wise Expenses:

Food : 600

Travel : 200

Shopping : 600

Other : 300

Highest Spending Category: Food / Shopping

Spending Status: Moderate Spending

ADVANCED LEVEL MINI PROJECT

Breathe for Today... Solve the advanced problem !